

YoungJo Choi (Jo Choi)

Jo Choi | Brooklyn, New York, USA | yc7330@nyu.edu | +1-312-918-9643 | **Github:** TechJoe96
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EDUCATION

New York University

- M.S., Computer Engineering

Fall 2024-Present
(Expected graduation date: May 2026)

Illinois Institute of Technology (Dual Degree program)

- B.S., Computer Engineering, minor in Artificial Intelligence

Fall 2022-Spring 2024

Inha University (Dual Degree program)

- B.S., Electronic Engineering

Spring 2016-Spring 2024

SKILLS

- **Programming&Scripting:** SystemVerilog, Verilog, VHDL, RISC-V Assembly, ARM Assembly, MATLAB, Python, C++, C, Java, PostgreSQL, MySQL, R
- **Design Tools:** Cadence Virtuoso/Genus/Innovus, Xilinx Vivado, ModelSim, KiCAD, AutoCAD
- **AI/ML Frameworks:** PyTorch, TensorFlow, Scikit-learn, Keras, MLflow
- **Data & Visualization:** Pandas, NumPy, PowerBI, Tableau, Excel, Selenium
- **Other:** React, HTML/CSS/JavaScript, Node.js, REST API, FastAPI, Flask, SageMaker AWS, Docker, Kubernetes, Linux, Git

WORK EXPERIENCE

Teaching Assistant at New York University

Sep 2025-Dec 2025

- Teaching lab sections for ECE-UY 3114 Fundamentals of Electronics I at New York University
- Ran experiments across amplifiers, diodes and transistor circuits

Data Analyst Intern at C&B Trading in Seoul, Korea

Jun 2024-Aug 2024

- Improved sales forecast accuracy to 90% with a predictive model; built Excel reports and PowerBI dashboards that identified three key process bottlenecks for senior management.

Republic of Korea (ROK) Military

Aug 2017-Apr 2019

- Honorably discharged as a Sergeant and a Squad Leader after serving in ROK for 21 months

RESEARCH EXPERIENCE & PROJECTS

DDR4-2400 SDRAM Memory Controller

Spring 2026

- Designed a JEDEC-compliant DDR4 controller with bank-group-aware scheduling, 16 per-bank state machines, timing enforcement, and auto-refresh with deferral (~5,100 lines, 13 RTL modules)
- Implemented protocol stack: power-on init (MRS/ZQCL), digital PHY (DQS/ODT), AXI4 slave frontend with burst expansion
- Verified with behavioral DRAM model and scoreboard — 10/10 tests pass, 0 data mismatches, 0 timing violations

OpenPOWER SoC Design with Microwatt CPU and NN Accelerator

Fall 2025

- FPGA/ASIC-ready neural network accelerator (Verilog RTL) featuring INT8 systolic array processing for OpenPOWER systems

GenAI-based Chip Design

Fall 2025

- 1) Developing LLM model architectures for automating chip design across RTL, verification to physical design, also for design optimization under the guidance of Prof. Ramesh Karri
- 2) Designing a Quantized Neural Net Accelerator with the Microwatt OpenPOWER CPU on SKY130, enabling efficient ML inference via memory-mapped I/O and custom instruction paths

Efficient Vision-Language Model (VLM) Design with Sparse Techniques

Sep 2025-Present

- 1) Developing a saliency-based VLM by integrating a Vision Transformer with an LLM through a projection layer, enabling efficient visual token selection and multimodal reasoning under the guidance of Prof. Sai Qian Zhang
- 2) Reproduced SparseVLM on LLaVA-1.5-7B, reducing visual tokens by 30% with minimal accuracy loss; outperformed three weight-pruning baselines on VQAv2

Superconducting Nanoribbon Device Fabrication

Mar 2025-Aug 2025

- Developed superconducting nanoribbon wires on HbN substrate at Nano Lab under the guidance of Prof. Davood Shahrjerdi
- Engineered the recipes of E-beam lithography, Deep Reactive-Ion Etching, and Buffered Oxide Etching processes in cleanroom environment

Real-Time Parkinson's Tremor & Dyskinesia Detector

Spring 2025

- Built a real-time embedded system on STM32L475 that detects Parkinson's tremor (3–5 Hz) and dyskinesia (5–7 Hz) using accelerometer data and FFT-based frequency analysis
- Implemented a signal processing pipeline (DC offset removal, Hann windowing, 256-point FFT) capturing 156 samples at 52 Hz in 3-second analysis windows with real-time LED feedback

LEADERSHIP EXPERIENCE

- Active member at **Triangle fraternity (Armour Chapter)**

Fall 2022-Spring 2024

- **President & Founder** of Korean Student Association at Illinois Institute of Technology

Summer 2023-Spring 2024